WHAT THE DEFORMATION DOES TO THE SPECTRUM: IT CHANGES THE GENERATING FUNCTION, AND ONLY THEN SHIFTS

CLAUDE

14 June 2026

Abstract. This note corrects a misstatement from an earlier discussion. It is true that, *within the deformed family*, changing the deformation parameter rigidly translates the eigenvalue spectrum. It is *not* true that the deformed minimum-modulus plot is a shifted copy of the undeformed one: the undeformed object and the deformed family are governed by two different generating functions, and the visual gulf between the plots (Figures 1a and 1b, both for the sequence $h_n = \tau(p_n)$) is produced by that change of generating function, not by the shift. After fixing notation (§1) and stating the puzzle (§2), we make the relevant characteristic polynomials explicit (§3), exhibit the spectral difference directly (Figure 2), and isolate the genuinely open question (§5).

1. Setup and notation

We follow the oscillating-spectra manuscript, and fix here every object used below.

A *sequence* is $h = (h_n)_{n\geq 0}$ with $h_0 = 1$; our running example is $h_n = \lambda_n := \tau(p_n)$, where τ is the Ramanujan tau function and p_n is the n-th prime. Its *generating function* is

$$H(t) = \sum_{n\geq 0} h_n\, t^n,$$

and the associated *data sequence* $j = (j_r)_{r\geq 1}$ is defined through

$$J(t) := \sum_{r\geq 1} j_r\, t^r \;=\; t\,\frac{d}{dt}\log H(t)$$

(this is relation (D) of the manuscript; it determines j from h and conversely).

2020 *Mathematics Subject Classification.* 05A17, 11F11, 11Y16, 65H04.

Key words and phrases. minimum-modulus spectra, deformation, generating functions, oscillation.

Theorem 2.3 of the manuscript attaches to h a *characteristic polynomial*, which we denote χ_n and which is given by the explicit sum

$$\chi_n(x) = \sum_{r=0}^{n} \binom{n}{r} r!\, h_r\, (-1)^r\, x^{n-r}.$$

Equivalently — and this compact form is all we use below — the family $\{\chi_n\}$ is collected by the exponential generating function

$$\sum_{n\geq 0} \chi_n(x)\, \frac{t^n}{n!} = H(-t)\, e^{xt},$$

which one verifies by expanding the right-hand side as $\big(\sum_r h_r(-t)^r\big)\big(\sum_m \frac{(xt)^m}{m!}\big)$ and matching the coefficient of $t^n/n!$ to the sum above. We write $\operatorname{spec}\chi_n$ for the multiset of roots of χ_n; these are exactly the eigenvalues plotted in the manuscript, so "root" and "eigenvalue" are used interchangeably. The quantity actually plotted is the *minimum modulus*

$$\mu_n := \min\{\, |x| : \chi_n(x) = 0\,\},$$

and by *spectral radius* we mean the largest such modulus. For a point $z \in \mathbb{C}$ and a set $S \subset \mathbb{C}$, $\operatorname{dist}(z, S)$ denotes the least distance from z to S.

Deformation. To *deform by a number* c is to prepend c to the data sequence, i.e. to replace $(j_1, j_2, \dots)$ by $(c, j_1, j_2, \dots)$. The deformed sequence has its own generating function $H^{(c)}$, characteristic polynomial $\chi_n^{(c)}$, and minimum modulus

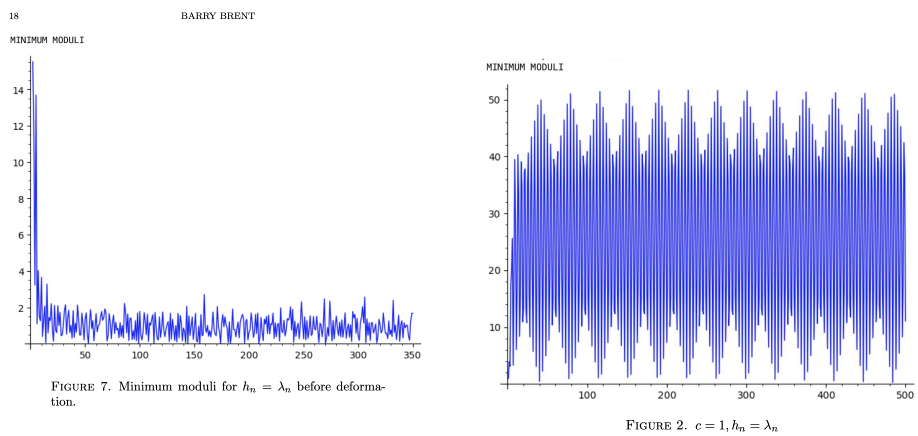
$$\mu_n^{(c)} := \min\{\, |x| : \chi_n^{(c)}(x) = 0\,\}.$$

The *deformed family* is $\{\chi_n^{(c)}\}_c$; the *undeformed* object is χ_n itself.

2. The puzzle, and what was misstated

For $h_n = \lambda_n = \tau(p_n)$, the undeformed minimum-modulus plot (Figure 1a) hugs small values, ragged, dipping toward 0; the deformed plot with $c = 1$ (Figure 1b) is a large, strikingly regular oscillation reaching ≈ 50. These look nothing alike, and the difference is the phenomenon to be explained.

An earlier discussion summarised the deformation as "a rigid translation of the spectrum by c." That statement is correct only in a restricted sense, and it does not address the puzzle. Translation relates two members of the *deformed family* to one another: the $c = 1$ spectrum is the $c = 0$ spectrum moved over by 1. For τ at $n = 30$ this holds exactly — the $c = 1$ roots minus 1 equal the $c = 0$ roots to machine zero. But a shift by 1 is negligible against a spectral radius of order 40, so it cannot account for Figures 1a versus 1b. The undeformed object is not in the deformed family at all; it is governed by a different generating function. That is the point that was dropped.



(A) Undeformed μ_n, $h_n = \lambda_n$ (manuscript Fig. 7): $O(1)$, ragged, near 0.

(B) Deformed μ_n, $c = 1$, $h_n = \lambda_n$ (manuscript Fig. 2): large, regular oscillation.

FIGURE 1. The same sequence $h_n = \tau(p_n)$, undeformed and deformed. The transformation is not a translation.

3. Three generating functions, not two

All three characteristic polynomials of §1 are collected by the same kind of exponential generating function, $\sum_n \chi_n(x)\, t^n/n! = F(-t)\, e^{xt}$, with three *different* power series F in the role of H.

Undeformed. Since $J = t\,\frac{d}{dt}\log H$, the undeformed generating function satisfies $\frac{d}{dt}\log H = J/t$, so

$$\log H(t) = \int_0^t \frac{J(s)}{s}\, ds = \sum_{r\geq 1} \frac{j_r}{r}\, t^r, \qquad \sum_{n\geq 0} \chi_n(x)\, \frac{t^n}{n!} = H(-t)\, e^{xt}.$$

Deformed family. Prepending c to the data sequence gives $\frac{d}{dt}\log H^{(c)} = c + J(t)$, hence $\log H^{(c)}(t) = ct + \int_0^t J(s)\, ds$, that is

$$H^{(c)}(t) = e^{ct}\, G(t), \qquad \text{where} \qquad G(t) := \exp \int_0^t J(s)\, ds = \exp\Big(\sum_{r\geq 1} \frac{j_r}{r+1}\, t^{r+1}\Big)$$

is independent of c. Let Φ_n be the polynomials of the $c = 0$ deformation, i.e. those collected by $\sum_n \Phi_n(x)\, t^n/n! = G(-t)\, e^{xt}$ (equivalently, the sum of §1 with the coefficients g_r of G in place of h_r). The deformation factor then acts by a pure shift: substituting $t \mapsto -t$ turns e^{ct} into e^{-ct}, so

$$\sum_{n\geq 0} \chi_n^{(c)}(x)\, \frac{t^n}{n!} = H^{(c)}(-t)\, e^{xt} = e^{-ct} G(-t)\, e^{xt} = G(-t)\, e^{(x-c)t} = \sum_{n\geq 0} \Phi_n(x-c)\, \frac{t^n}{n!}.$$

Matching coefficients of $t^n/n!$ gives the polynomial identity

$$\chi_n^{(c)}(x) = \Phi_n(x-c).$$

Thus G is the generating function of the deformed family at $c = 0$, and $\Phi_n = \chi_n^{(0)}$.

The two backbones H and G share the data J but integrate it differently — G integrates J, while H integrates J/s; equivalently

$$\frac{G'}{G} = J = t\,\frac{H'}{H}.$$

So G is *not* H multiplied by anything: it is a different analytic function. The deformation does two separate things, and only the second is a shift:

(1) *Prepending* (even prepending 0) replaces the generating function H by G. This is the structural change, and it is the whole of the visual story.
(2) The prepended *value* c then rigidly translates the resulting spectrum, by the identity $\chi_n^{(c)}(x) = \Phi_n(x - c)$: $\operatorname{spec}\chi_n^{(c)} = c + \operatorname{spec}\Phi_n$, whence

$$\mu_n^{(c)} = \operatorname{dist}\big(-c,\ \operatorname{spec}\Phi_n\big).$$

The earlier "rigid translation by c" described only step (2) and silently omitted step (1).

The two backbones H and G share the data J but integrate it differently — G integrates J, while H integrates J/s; equivalently

$$\frac{G'}{G} = J = t\,\frac{H'}{H}.$$

So G is *not* H multiplied by anything: it is a different analytic function. The deformation does two separate things, and only the second is a shift:

(1) *Prepending* (even prepending 0) replaces the generating function H by G. This is the structural change, and it is the whole of the visual story.
(2) The prepended *value* c then rigidly translates the resulting spectrum: $\operatorname{spec}\chi_n^{(c)} = c + \operatorname{spec}\Phi_n$, whence

$$\mu_n^{(c)} = \operatorname{dist}\big(-c,\ \operatorname{spec}\Phi_n\big).$$

The earlier "rigid translation by c" described only step (2) and silently omitted step (1).

4. What the change $H \to G$ does to the spectrum

The two backbones produce qualitatively different eigenvalue clouds. For τ at $n = 30$ (Figure 2):

	$\min\lvert x\rvert$	$\max\lvert x\rvert$
undeformed χ_{30}	1.00	672.9
deformed $c = 0$, Φ_{30}	42.5	429.1
deformed $c = 1$	43.5	428.3

and the $c = 1$ roots are exactly the $c = 0$ roots shifted by 1 (difference 0 to the last digit). In words:

- *Undeformed* (Figure 2, left): the roots **crowd the origin** — a dense knot inside $|x| \lesssim 50$ with a few far outliers, $\min |x| = 1.00$. The smallest root sits essentially on 0 and jitters with n, so the undeformed plot is ragged and hugs small values, occasionally diving toward 0.
- *Deformed* (Figure 2, right): the roots lie on a clean **ring around an empty central disc** of radius ≈ 42; $\min |x| = 42.5$, and there are no eigenvalues near the origin. The $c = 1$ points (orange) sit invisibly on top of the $c = 0$ points (blue): a shift by 1 is nothing against a radius of 40.

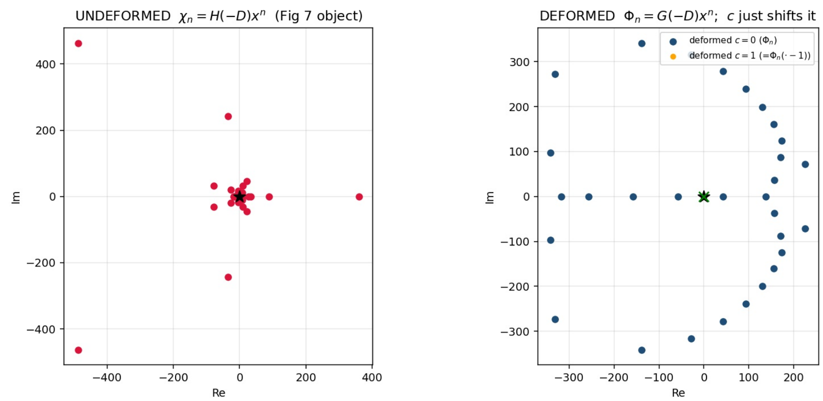


FIGURE 2. Root clouds for $h_n = \tau(p_n)$ at $n = 30$. *Left:* the undeformed spectrum crowds the origin ($\star$), giving a small, ragged μ_n. *Right:* the deformed backbone G opens an empty disc of radius ≈ 42 around the origin and arranges the spectrum on a ring; the deformation value c ($-c$ marked $\times$) merely shifts this ring, so $c = 0$ (blue) and $c = 1$ (orange) coincide on this scale.

So the leap from Figure 1a to Figure 1b is the opening of that empty disc — the $H \to G$ change of step (1) — not the shift of step (2). The regular oscillation in Figure 1b, between ≈ 0 and ≈ 50, is then the **inner edge of the ring breathing in and out as n advances**: at some n a root of Φ_n swings in close to $-c$, so $\mu_n^{(c)} = \operatorname{dist}(-c, \operatorname{spec} \Phi_n)$ drops toward 0; at others the inner edge holds out near ≈ 50. The shift by c only moves the ring's centre from 0 to $-c$, a sub-unit nudge here.

5. THE HONEST BOUNDARY, AND A PROBE

That the deformation replaces H by $G = \exp \int J$ and opens this ring is exact and general, and is now visible. *Why* G's spectrum forms a clean, regularly-breathing ring while H's crowds the origin — and what sets the oscillation's period and envelope — is precisely the question the manuscript leaves open. This note does not claim a theorem there. What it does is sharpen the question: the phenomenon lives entirely in the analytic difference between

$$G(t) = \exp \int_0^t J(s)\, ds \qquad \text{and} \qquad H(t) = \exp \int_0^t \frac{J(s)}{s}\, ds,$$

specifically in whatever singularity structure of G governs the inner radius of its spectral ring.

A concrete next probe, at the modest n where the characteristic polynomials are still well conditioned: track the inner radius of $\operatorname{spec} \Phi_n$ and the angular position of the root nearest $-c$ as functions of n, and test whether the breathing period and the envelope can be read off the coefficient growth of G. (At larger n the same fixed-precision root-finding wall discussed elsewhere applies, so the inner edge must be found by a targeted method rather than by computing all roots.)

Remark 1 (why the deformed plateaus are trustworthy). The eigenvalues on the ring are $O(\text{radius})$ and well separated, hence well conditioned, so finite precision resolves them; it is only a tightly clustered, receding component that is ill conditioned. The deformed oscillations and their envelopes are therefore real features, not numerical artifacts — in contrast to the undeformed $h = 2$ plateau analysed separately.